

Takahiro Hirano

+81-90-2847-5613 // takahiro.hirano@gmail.com // https://github.com/takahiro-hirano

EDUCATION

Tokyo Institute of Technology, *B.S. Computer Science*

April 2022 - March 2026

- Completed coursework in Machine Learning, Deep Learning, and Probability & Statistics with 3.8/4.0 GPA
- Led study group of 8 students in Advanced Algorithms course, achieving 92% average exam performance

EXPERIENCE

Preferred Networks, *Machine Learning Intern*

June 2024 - August 2024

- Implemented and benchmarked 3 different convolutional neural network architectures on CIFAR-10 dataset, achieving 94.2% accuracy with ResNet-18 variant
- Optimized PyTorch model inference pipeline using quantization techniques, reducing inference latency by 38% on edge devices
- Collaborated with 2 senior engineers to debug gradient flow issues in custom transformer layer, reducing training loss convergence time from 8 hours to 4.5 hours

PROJECTS

Sentiment Analysis on Japanese Tweets, *Python, BERT, PyTorch, Hugging Face, scikit-learn*

February 2024 - March 2024

- Built fine-tuned BERT-based classifier for Japanese language sentiment detection with 87% F1-score on 5,000 labeled tweet samples
- Developed data preprocessing pipeline using tokenization and augmentation techniques that increased training dataset diversity by 40%

Hackathon: Image Classification for Medical X-Rays, *TensorFlow, Keras, NumPy, Matplotlib*

November 2023

- Engineered EfficientNet B0 model trained on 2,048 labeled chest X-ray images, achieving 89.5% AUC-ROC on holdout test set
- Ranked 3rd place among 42 teams by implementing class balancing and cross-validation strategies that improved model generalization

Time Series Forecasting: Stock Price Prediction, *LSTM, TensorFlow, Pandas, Matplotlib, yfinance API*

December 2023 - January 2024

- Constructed multi-layer LSTM neural network to forecast daily closing prices for 5 major stocks with MAPE of 3.2%
- Engineered feature engineering pipeline extracting 12 technical indicators (RSI, MACD, Bollinger Bands) improving model accuracy by 15%

EXTRACURRICULAR

Tokyo Tech AI Club, *Core Member*

April 2023 - Present

- Organized and conducted 6 monthly workshops on ML fundamentals, attracting 50+ undergraduate participants per session
- Mentored 4 junior students on implementing neural networks from scratch, with 100% project completion rate

Kaggle Competition Platform, *Active Competitor*

September 2023 - Present

- Competed in 8 machine learning competitions with top-15% finish in 3 competitions through ensemble methods and hyperparameter optimization
- Earned 892 Kaggle discussion upvotes by sharing reproducible code walkthroughs and feature engineering insights

TECHNICAL SKILLS

Languages: Python, SQL, Japanese (Native), English (Fluent)

Technologies: PyTorch, TensorFlow, Keras, scikit-learn, Hugging Face, BERT, CNNs, RNNs, LSTMs, XGBoost

Tools: Jupyter Notebook, Git, GitHub, Weights & Biases, Colab, Docker, Linux